

A **50 kW electrical (kWel) biomass wood chip CHP (Combined Heat and Power) unit** typically operates via small-scale wood gasification. This highly efficient process converts raw wood chips into a clean, combustible syngas to drive an internal combustion engine coupled to a 50 kW generator. [1, 2, 3, 4, 5]

Typical Technical Specifications

For a commercial 50 kWel wood chip CHP unit, you can expect the following technical baselines: [6, 7]

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- **Electrical Output:** ~45 to 56 kWel (alternating current, ready for grid feed-in or on-site use).
- **Thermal Output:** ~95 to 130 kWth (available as hot water up to 90°C for district heating, kilns, or drying systems).
- **Total Efficiency:** ~85% to 90% (leveraging both electricity and recaptured thermal energy).
- **Fuel Consumption:** Roughly 40 to 50 kg of wood chips per hour of continuous operation. [7, 8, 9, 10, 11, 12]
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Strict Fuel Requirements for Wood Chips

Unlike large-scale biomass boilers, small-scale 50 kW gasification systems are highly sensitive to fuel quality. For reliable operation without tar buildup, the wood chips must meet strict specifications: [13, 14]

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- **Moisture Content:** Crucially low, typically between **8% and 15%** water content. Standard "green" or fresh wood chips (~40-50% moisture) must undergo a pre-drying process before entering the gasifier.
- **Chip Size:** Highly uniform sizing—most commonly **P31S or P45 standards** (roughly 30–50mm).
- **Fines & Ash:** Low dust/fine material content to prevent system clogging and an ash content below 1%. [6, 7, 13, 15, 16]
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Top Commercial Manufacturers

Several specialized European companies offer turn-key, containerized 50 kW wood chip CHP solutions: [6]

1. [Spanner Re² \(HKA 50\)](#): Generates 49 kWel and 106 kWth. They offer comprehensive, fully automated turn-key setups from the wood chip dryer to the gasifier.
2. [Wegscheid Entrenco \(WE-50C\)](#): Rates at 50 kWel and 120 kWth, explicitly designed for agriculture, forestry, and wood processing plants.
3. [Fröling \(CHP 50\)](#): An autothermal fixed-bed downstream gasifier offering a 46-56 kWel output range inside a "plug and play" container.
4. [Volter \(Walter Powerplant\)](#): High-end indoor or containerized option yielding 50 kWel and 130 kWth.
5. [Glock Eco Energy \(CHP 50\)](#): Austrian-engineered automated off-grid or high-demand solution delivering 50-57 kWel. [6, 7, 8, 9, 17, 18, 19]

Ideal Use Cases & Economics

These systems are ideal for decentralized, baseload power generation where a year-round, matching demand for heat exists. Common deployments include: [14, 15]

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- **Sawmills and Woodworking Shops:** Utilizing their own dry wood residues as free fuel.
- **Agriculture & District Heating:** Powering local grids while using waste heat to dry crops, warm stables, or heat a village district.
- **Hotels & Wellness Centers:** Facilities with massive, constant thermal needs for pools and hot water. [6, 15, 20, 21, 22]
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To help provide more specific information, please share:

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- What is your **intended application** (e.g., farm, sawmill, remote off-grid facility)?
- Do you have a **ready source of wood chips**, or will you need to source and dry them?
- Do you plan to **use the heat locally**, or are you primarily focused on generating electricity?
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[1] <https://www.ruralenergy.co.uk>

[2] <https://www.powermaxgasifiers.com>

[3] <https://www.inf.gov.nt.ca>

[4] <https://www.sciencedirect.com>

[5] <https://burkhardt-gruppe.de>

[6] <https://we-bioenergy.com>

[7] <https://re2.energy>

[8] <https://www.froeling.com>

[9] <https://volter.fi>

[10] <https://www.syncraft.at>

[11] <https://chpalliance.org>

[12] <https://cdn.forestresearch.gov.uk>

[13] <https://www.treco.co.uk>

[14] [<https://projectblue.blob.core.windows.net>]

(<https://projectblue.blob.core.windows.net/media/Default/Horticulture/GrowSave/31 - Biomass CHP.pdf>)

[15] <https://www.youtube.com>

[16] <https://www.sciencedirect.com>

[17] <https://www.commercialbioenergy.ca>

[18] <https://re2.energy>

[19] <https://www.froeling.com>

[20] <https://we-bioenergy.com>

[21] <https://www.youtube.com>

[22] <https://woodybpower.biz>